

Data Science Report

Data Cleaning

Column Name	Situation	Action	Justification
Brand	Satisfactory	Merged	These will only be for labelling laptops, not comparison, so they were combined.
Model			
Screen Size	Stored as non-numerical	Made numerical	Numerical data allows numerical analysis.
Hard Disk			
RAM			
Price			
Colour	Irrelevant	Dropped	Colour is irrelevant for a laptop's usability and recommendation, it's not a functional consideration.
Special Features	34% Missing values, listed data		Missing data causes unreliability. The available information often lists basic features, which might be present in laptops not explicitly listing them, biasing analysis.
CPU Speed	87% Missing values		Missing data causes unreliability. Relevant CPU information can be derived from the CPU column for more reliable comparisons.
Rating	60% Missing values		Data is largely missing; ratings, being subjective, are prone to bias. Questionable reliability due to the unknown number of ratings per entry.
CPU	Excessive levels	Grouped together similar levels	Values usually belong to a series with similar properties, grouping simplifies comparison. Acknowledging rarely occurring values in a group recognizes additional options, giving them value.
OS			
Graphics			
Graphics Coprocessor			
Hard Disk	Inconsistent format	Standardised units	Numerical data is stored with a uniform unit for consistency.
RAM			
Model		Lowercased, removed whitespace	Easier to compare categorical data with a consistent format.
CPU			
OS			

Graphics			
Graphics Coprocessor			
Screen Size	Missing values	Mode-filled missing values	With a small standard deviation and missing value count, the high-frequency mode suggests missing values may align with it with minimal impact. Figures 1 & 2 highlight this.
RAM			
Graphics	Conflicts with Graphics Coprocessor	Repositioned some values to Graphics Coprocessor	Now indicates whether graphics are integrated or dedicated, most data was already in this form, other values exist in Graphics Coprocessor. This information is also important for recommendations.
Graphics Coprocessor	60% Missing values, conflicts with Graphics	Merged values from Graphics	It now uniquely stores the graphics card or coprocessor, consolidation of common information simplifies data retrieval.
Price	Outliers	Removed outliers	Outliers exceeding 5x the interquartile range above the upper quartile would significantly bias the data, Figure 3 displays this.
All	Duplicate rows	Removed duplicates	Duplicate rows are redundant and can skew data.

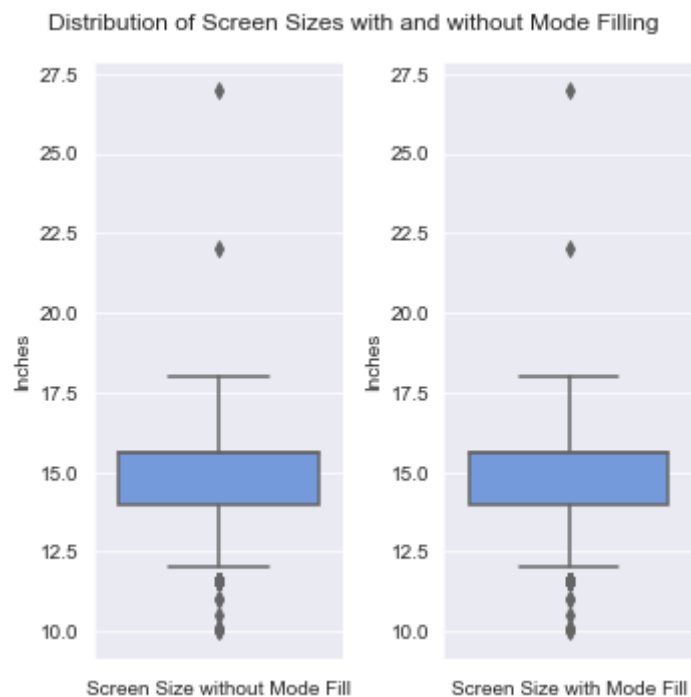


Figure 1 The impact of mode filling on the screen sizes

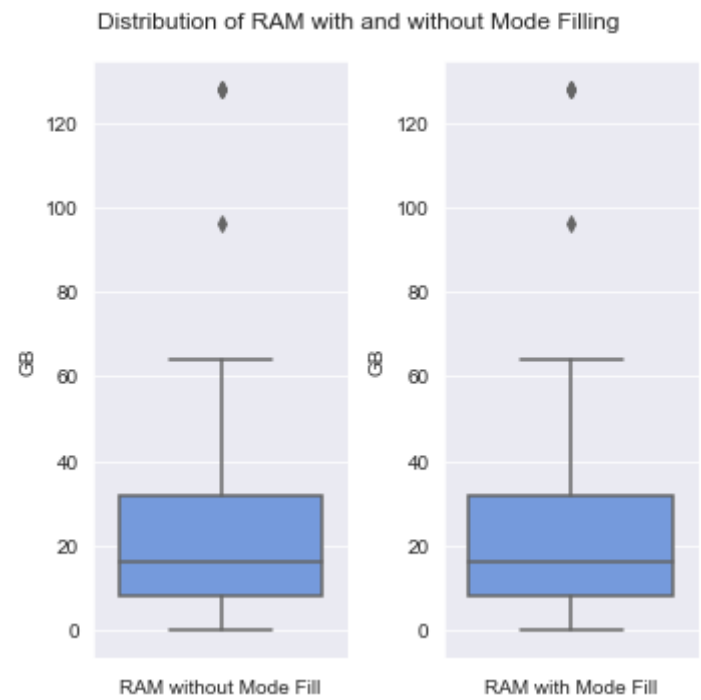


Figure 2 The impact of mode filling on RAM

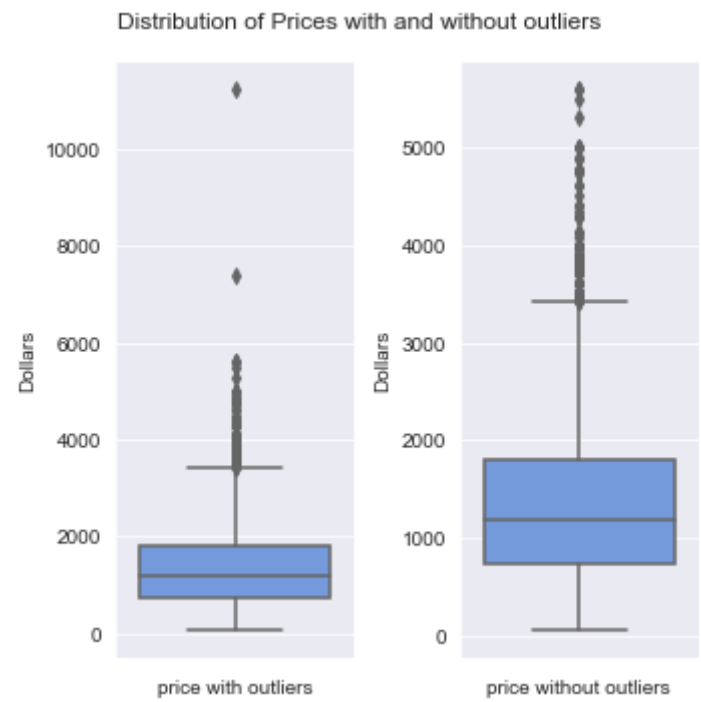


Figure 3 The effect of outlier culling on prices

Data Analysis

Humanities students likely prefer affordable, portable laptops with high storage that prioritise battery life over processing power, favouring smaller screens and lower processing power. Gamers prioritise processing and graphics capabilities, and larger screens, placing less emphasis on price and storage.

Word Count: 748

Figure 4 suggests there's a correlation between specification quality, generally improvements in one component accompany improvements in others. While this is optimal, it also implies fewer options for laptops with less processing power but high storage.

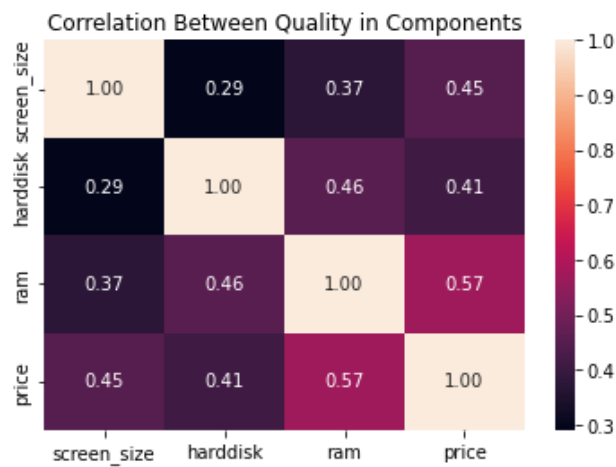


Figure 4 The correlation between numerical components

Larger screens, more RAM, and higher disk space correlate with higher prices and reduced laptop availability. For affordability and portability, screens of ≤ 14 inches are optimal averaging \$690. RAM of ≤ 8 GB is cost-effective (average \$641), while 1000 GB storage strikes a balance between price (around \$1000) and availability (627). Larger RAM and disk space options have significantly less availability (111) and higher prices (average \$1055).



Figure 5 The relationship of components against price

Intel Core i5 processors are the most common (623 laptops) and 86% of laptops have integrated graphics (better battery life). Intel Core i7 closely follows with a higher average price of \$839, however i5 suits lower-end preferences, Celeron, averaging \$305, is a cheaper alternative. Higher-end processors like Ryzen 9 or Core i9 are limited and expensive; therefore, Core i7, with the largest proportion of dedicated graphics (24%), is a potential substitute.

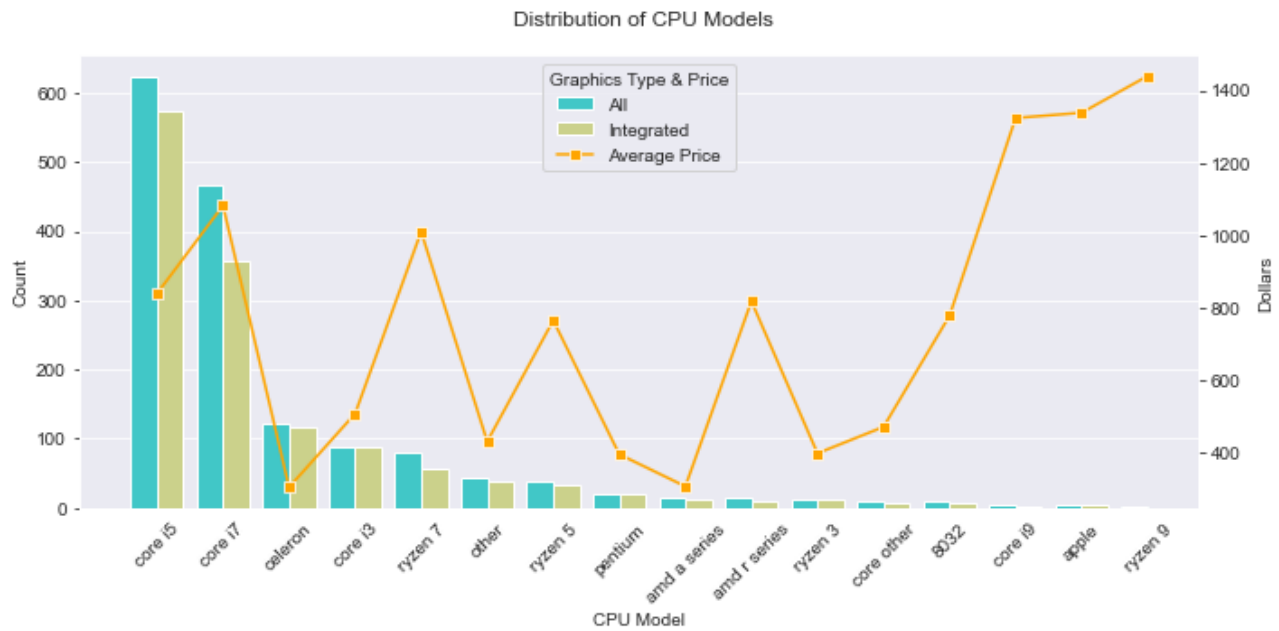


Figure 6 The distribution of CPUs and graphics type

Best laptops for humanities students: **Dell Latitude 3330, Dell Latitude 5320 2-in-1, Dell Latitude 5320, Asus Zenbook S, Dell XPS 9300**. These budget-friendly laptops have low-to-mid-range CPUs, integrated graphics for extended battery life, small screens (~13.5 inches) for portability, and ample storage (~1 TB).

Best laptops for gamers: **Asus Vivobook Pro 15, Shoxlab m15r5, MSI Katana 15, Asus Vivobook Pro 16, MSI Katana 17 b13vfk-835us**. These laptops have high-performance Core i9, Ryzen 9, or Core i7 CPUs, dedicated Nvidia GeForce RTX graphics cards, and 16 or 32 GB RAM.

References

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